ME 207: Fluid Dynamics

Lab Experiment 1: The Pitot tube

Date of Experiment: - 19/1/2025
Date of Submission: - 26/1/2025

Group 14

Atharva Suryarao	23110051
Khandvi Parth Piyushbhai	23110167
Kedar Kanzariya	23110156
Jainil Maru	23110200
Paras Shirvale	23110232

Abstract

The aim of the experiment is to study and verify Bernoulli's equation using the pitot tube setup by measuring velocities by two different methods. The conditions and external phenomena are also touched upon while analyzing the results.

1 Task 1.1

Q1: Read about Bernoulli's equation in Fluid Mechanics. Fundamentally, what is Bernoulli's equation? What are the assumptions under which this equation is valid?

The Bernoulli equation is a concise equation that focuses on conserving the kinetic, potential, and flow energies of a fluid stream and their conversion to each other. It gives us an approximate relationship between pressure, velocity, and elevation and is valid only in regions of steady, incompressible flow where net frictional forces are negligible. A key approximation in its derivation is neglecting viscous effects compared to other gravitational and pressure effects. Since all fluids have viscosity (though minor), we cannot apply the Bernoulli equation everywhere in a flow. However, the approximation is reasonable in regions such as inviscid flow regions, and we can apply the Bernoulli equation.

Q2: Define Static pressure and Stagnation pressure in a flow. How can these two be used to determine the velocity?

The Bernoulli equation states that the sum of the flow, kinetic, and potential energies of a fluid is constant. Therefore, the kinetic and potential energies can be inter-converted to flow energy and vice versa, causing the pressure to change. It can be better understood by multiplying the Bernoulli equation by the density (ρ):

$$P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

Each term in this formula has pressure units, and thus each term represents some kind of pressure:

- P is the **static pressure**, which is the actual thermodynamic pressure of the fluid. It is the same pressure as the one obtained from the thermodynamic property table.
- $\frac{1}{2}\rho v^2$ is the **dynamic pressure**; it represents the increase in pressure when the fluid in motion is isentropically stopped.
- The sum of static pressure and dynamic pressure is called **stagnation pressure** and is mathematically expressed as:

$$P_{stag} = P + \frac{1}{2}\rho v^2$$

.

These two pressures can be used to determine the velocity of the fluid. Rearranging the stagnation pressure equation for the velocity of the fluid gives:

$$v = \sqrt{\frac{2(P_{stag} - P)}{\rho}}$$

Q3: What is head loss in pipe flows (or internal flows)? Why is it important? Can you quantify the head loss?

In an analysis of pipe systems, pressure losses are generally expressed in terms of an equivalent fluid height, called **head loss** (h_l). The loss of head h_l represents the additional height that a pump must raise to overcome the friction losses in a pipe. These losses are caused by the viscosity of the fluid, which is directly proportional to the wall shear stress.

Relationship with Pressure Loss
From the statics of the fluid, we know the relationship between pressure and fluid height:

$$\Delta P = \rho g h$$

where:

- ΔP = Pressure drop in the pipe (Pa),
- ρ = Density of the fluid (kg/m^3),
- g = Acceleration due to gravity (m/s^2),
- h = Fluid height (m).

Rearranging for h, the fluid height (or head loss) is given by:

$$h_l = \frac{\Delta P}{\rho g}$$

Why is Head Loss Important?
Head loss is critical to understand for several reasons:

1. **Pressure Reduction:** It reduces the pressure of fluid flowing through the pipe. Without understanding the extent of pressure loss, it is impossible to accurately predict the system's behavior.
2. **Efficiency of Flow:** Head loss directly impacts the efficiency of fluid transport through the pipe.
3. **Power Requirements:** To compensate for head loss, additional power is required to pump the fluid. Quantifying head loss allows engineers to estimate the energy or power needed to maintain desired flow rates.

Quantifying Head Loss
Head loss in pipes can be divided into two categories:

1. **Major Losses:** Caused by friction along the length of the pipe. It can be quantified using the Darcy-Weisbach equation:

$$h_f = f\frac{L}{D}\frac{v^2}{2g}$$

where:

- h_f = Frictional head loss (m),
- f = Darcy friction factor (dimensionless),
- L = Length of the pipe (m),
- D = Diameter of the pipe (m),
- v = Flow velocity (m/s).

2. **Minor Losses:** Caused by fittings, bends, valves, etc., and can be expressed as:

$$h_m = K\frac{v^2}{2g}$$

where:

- h_m = Minor head loss (m),
- K = Loss coefficient (dimensionless).

Q4: What is the kinetic energy coefficient in an internal flow? Can the Bernoulli's Equation be modified to account for non-uniform velocity profile and head losses due to viscosity?
A kinetic energy coefficient is a correction factor used for the non-uniform velocity profile. It allows us to use the average velocity in the kinetic energy formula to find the kinetic energy at the cross-section. The average velocity is used because the velocity at any point in the cross-section differs, so the total kinetic energy differs. So, the product of the kinetic energy coefficient and the kinetic energy obtained from the average velocity equals the actual kinetic energy at the cross-section. It is also represented as the ratio of the actual kinetic energy to kinetic energy obtained by the average velocity.

$$\int_A \frac{V^2}{2}\rho V\,dA = \alpha\int_A \frac{\bar{V}^2}{2}\rho V\,dA = \alpha\dot{m}\frac{\bar{V}^2}{2}$$

$$\alpha = \frac{\int_A \rho V^3\,dA}{\dot{m}\bar{V}^2}$$

Where,

- α = Kinetic energy coefficient (Dimensionless),
- v = Flow velocity (m/s),

- ρ = Density of the fluid (kg/m^3),
- $\bar{V}$ = Average velocity (m/s),
- $\dot{m}$ = mass flow rate of the fluid (kg/s),
- A = cross-sectional area (m^2).

For laminar flow in the pipe, α=2.0.
In the turbulent flow, the velocity profile is flat. The kinetic energy coefficient can be obtained by the following equation.

$$\alpha = \frac{U^3}{\bar{V}^3} \cdot \frac{2n^2}{(3+n)(3+2n)}$$

Where,

- α = Kinetic energy coefficient $(Dimensionless)$,
- U = maximum velocity (m/s),
- $\bar{V}$ = Average velocity (m/s),
- n = power-law exponent $(Dimensionless)$.

$\frac{\bar{V}}{U}$ is a function of the power-law exponent n. The turbulent flow results lie in the range of n, from n=6 to n=10 for high Reynolds numbers. Hence, the α varies from 1.08 to 1.03 because α is nearly unity for the high Reynolds numbers and because the change in kinetic energy is small compared to the other dominant terms in the energy equation. We can always take the $\alpha = 1$ approximated value for the pipe flow calculations.

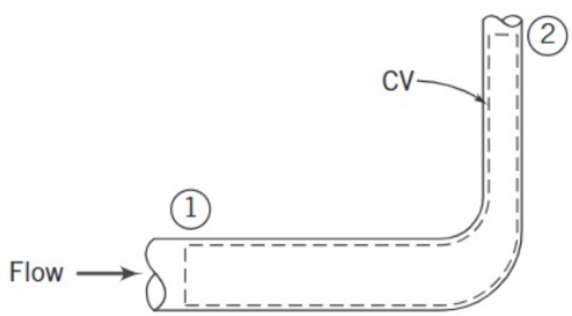

Figure 1: Control Volume

Modified Bernoulli equation:-

$$(\frac{p_1}{\rho} + \alpha_1 \frac{\bar{V_1}^2}{2} + gz_1) - (\frac{p_2}{\rho} + \alpha_2 \frac{\bar{V_2}^2}{2} + gz_2) = (u_2 - u_1) - \frac{\delta Q}{dm}$$

$$(\frac{p_1}{\rho} + \alpha_1 \frac{\bar{V_1}^2}{2} + gz_1) - (\frac{p_2}{\rho} + \alpha_2 \frac{\bar{V_2}^2}{2} + gz_2) = h_{l_T}$$

Where,

- ρ = Density of the fluid (kg/m^3),
- $u_2 - u_1$ = unwanted thermal energy (J/kg),
- h_{l_t} = Total head loss (m^2/s^2).

At the section-1,

- α_1 = Kinetic energy coefficient at the section-1 ($Dimensionless$),
- p_1 = pressure at the section-1 (Pa),
- $\bar{V_1}$ = Average velocity at the section-1 (m/s),
- $gz1$ = Gravitational potential energy per unit mass ($m2/s2$),

At the section-2

- p_2 = pressure at the section-2 (Pa),
- $\bar{V_2}$ = Average velocity at the section-2 (m/s),
- α_2= Kinetic energy coefficient at the section-2 ($Dimensionless$),
- gz_2 = Gravitational potential energy per unit mass (m^2/s^2),
- δQ = Heat (J),
- dm = infinitesimal mass of the fluid (kg).

$\frac{\bar{V}}{U}$ is a function of the power-law exponent n. The turbulent flow results lie in the range of n, from n=6 to n=10 for high Reynolds numbers. Hence, the α varies from 1.08 to 1.03 because α is nearly unity for the high Reynolds numbers and because the change in kinetic energy is small compared to the other dominant terms in the energy equation. We can always take the α=1 approximated value for the pipe flow calculations.

2 Introduction

Figure 2: Image of the experimental setup.

A pitot tube is often used to measure fluid velocity by taking advantage of the way kinetic energy turns into pressure differences. The device works based on Bernoulli's principle, which says that in steady, smooth-flowing fluids, the total pressure along a streamline stays the same. This total pressure consists of static pressure, dynamic pressure, and hydrostatic pressure. The pitot tube specifically captures the difference between static and dynamic pressures to determine the fluid's speed.

The pitot tube is a bent tube with an opening at its end that faces the fluid flow. This opening connects to a manometer, which measures the pressure difference between the stagnation point (where the fluid is stopped) and the static pressure of the moving fluid. The stagnation pressure is the highest pressure you get when the fluid is brought to rest, while static pressure is the pressure of the fluid as it moves. The difference between these two pressures, called dynamic pressure, is directly linked to the fluid's velocity.

Using a pitot tube, you can compare the actual fluid velocity with the theoretical velocity predicted by Bernoulli's equation. This offers a practical way to test and verify the principle in real-world conditions. The steps for carrying out this experiment are outlined in the next section.

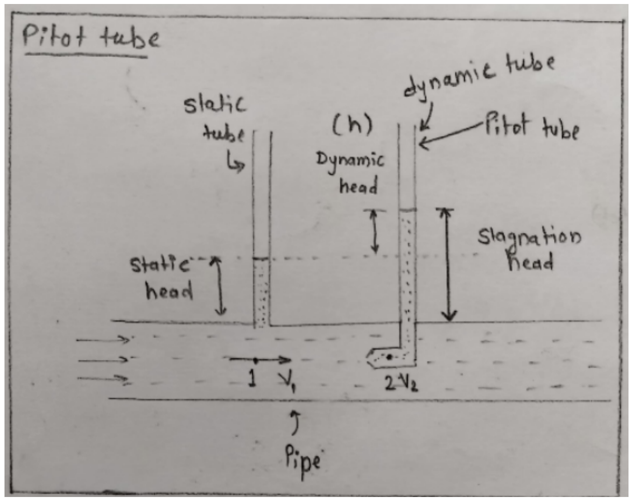

Figure 3: Pitot tube

3 Objectives

(i) Study the difference between static and stagnation pressures for various flow rates.
(ii) Determine the velocity from the Bernoulli's equation.
(iii) Interpret the results and the potential discrepancies between the theoretical analysis and the observed data.

4 Experimental Procedure

- First, we fix the height of the water in the constant head tank.
- A change in the water's height indicates a change in the flow rate. There were two valves to fix the height to constant.
- One valve is under the table; if we move this valve in a clockwise direction, it is closed, and the height of the water increases in the constant head tank, and if we move this valve in an anti-clockwise direction, it is opened. The height of the water decreases in the constant head tank.
- After fixing the water's height in the constant head tank, we noted the height values of the static and stagnation tubes.
- The static and stagnation pressure from the $P = \rho g h$ is different because the heights are different.
- Then, there is another valve in the pitot tube apparatus. If we move this valve clockwise, then the valve is closed, and the water will come into the burette.
- From this, we must measure the time until it reaches 5 cm. We note the measured time.

- During the time measurement, if the bubble occurs in the burette, we have to restart to measure the time (which means we first empty the burette and then record the time).
- We repeat this whole process for three different flow rates.

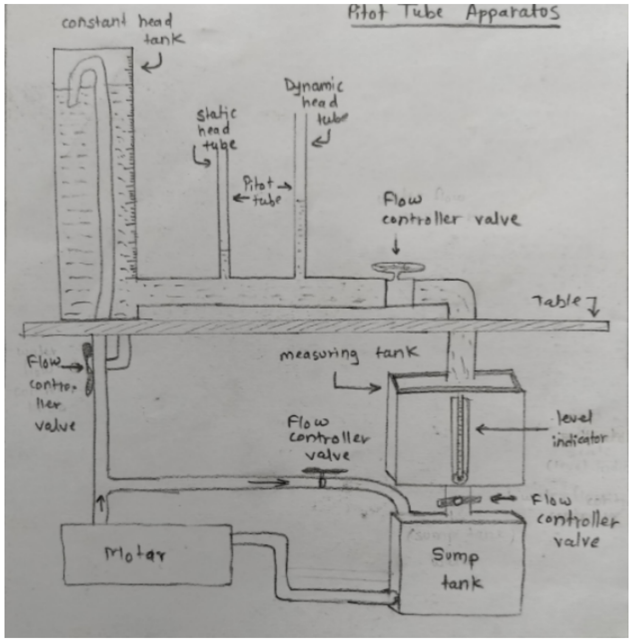

Figure 4: Schematic diagram of the experimental setup.

5 Results

5.1 Using the bernoulli's head equation

The equation mentioned in Q2, which relates the velocity with the stagnation pressure and the static pressure, can be reduced to the equation demonstrated below.

$$v = \sqrt{2g(h_2 - h_1)}$$

where:

- h_2 = Elevation in column corresponding to Stagnation pressure (m),
- h_1 = Elevation in column corresponding to Static pressure (m),
- g = Acceleration due to gravity (m/s^2),
- v = Flow velocity (m/s).

By measuring the values of h_1 and h_2, the velocity can be calculated using the expression demonstrated above.

Upon performing the experiment, the following velocity values were found:

Constant head (mm)	h_2 **(mm)**	h_1 **(mm)**	v **(m/s)**
400.0 ± 0.5	60.0 ± 0.5	43.5 ± 0.5	0.56 ± 0.017
370.0 ± 0.5	22.0 ± 0.5	9.0 ± 0.5	0.50 ± 0.019
451.0 ± 0.5	147.0 ± 0.5	134.5 ± 0.5	0.49 ± 0.019

Table 1: Values of velocity calculated using Head equation

5.2 Using the time required to fill the burette partially.

Apart from this, the velocity calculation was performed using the time it takes to fill the burette to the level of 5 cm.
The cross-sectional area of the burette was $0.16m^2$
Thus, the velocity of the flow is given by:

$$V = \frac{Ah}{\pi(r^2)t}$$

where:

- V = flow velocity,
- A = Cross-sectional area of burette,
- h = elevation of water level in burette,
- r = radius of the pipe,
- t = time taken for water to reach the elevation h.

Here, A=0.16 m^2, h=0.05m, r=0.02m

This gives the velocity as a function of time as shown below.

$$V = \frac{25.47}{t}$$

The following values of velocities were found to correspond to the different values of the time.

t (s)	**V** (m/s)
48.13 ± 0.01	0.5291 ± 0.000110
47.66 ± 0.01	0.5344 ± 0.000112
55.83 ± 0.01	0.4562 ± 0.0000817

Table 2: Values of velocity calculated using burette method.

The error in the values of velocities shown above is very small since human error is not included in the calculation because it is not quantifiable. The only error that constitutes the total error is the least count error, which is not significant in comparison to human error.

5.3 Comparison of values achieved using both the methods.

The table shown below compares the values of velocities measured by both methods.
It can be seen that the velocity calculated using the manual method approximately matches that of one calculated from the Bernoulli's head equation.

v (m/s)	**V** (m/s)
0.56 ± 0.017	0.5291 ± 0.000110
0.50 ± 0.019	0.5344 ± 0.000112
0.49 ± 0.019	0.4562 ± 0.0000817

Table 3: Comparison of velocities calculated through both the methods.

6 Error Analysis

6.1 Calculation of error

Errors in the measurement of all the quantities are calculated using the upper bound estimate.
It can be mathematically given as:

$$\sigma_Y = \sum_{i=1}^{n} \left| \frac{\partial f}{\partial x_i} \right| \sigma_i$$

Thus, the error in the value of velocity(v), calculated using Bernoulli's equation, is given as:

$$\sigma_v = \frac{g}{\sqrt{2g(h_2 - h_1)}}(\sigma_{h_2} + \sigma_{h_1})$$

Here, the fluctuation in columns was significantly less in comparison to the least count error.
Thus, $\sigma_{h2} = 0.5mm$ and $\sigma_{h1} = 0.5mm$

Similarly, the formula for calculating errors in the velocity values by thee manual method can be given as:
We can use this equation to calculate the velocity of each and every value that needs to be calculated.

$$\sigma_V = \frac{25.47}{t^2}\sigma_t$$

The error in the calculation of the velocity using the manual method is only dependent upon the time, which is calculated using the digital stopwatch. Thus, systematic errors can be neglected, and the total error in measuring time is given as the least count error, which is 0.1 seconds.
In this case, the total error is given as the least count error, but it is not an accurate assumption since there would be some epistemic uncertainties that can not be determined.

6.2 Reasoning

As observed in the result, the theoretical values of velocity approximately match those of the experimental one. However, there are slight deviations that can occur because of several factors, most of which are described in the first section. The following points can be taken care of in order to get better results.

- The number of readings can be increased, which can lead to more precise results and a reduction in the error.
- Unquantifiable errors, such as epistemic uncertainties or human errors, can be reduced by performing the experiment with more care.

7 Observations

The following observations were made after doing the analysis of the pitot tube experiment.

- Bernoulli's equation holds true and is valid for fluids with some assumptions like negligible viscosity and no surface tension between the walls of the pipe and fluid.
- There are deviations in the values of velocity, calculated using two different methods.
- Elevation in the level of water in constant head tank lead to the elevation in columns corresponding to the static pressure and the stagnation pressure.

8 Conclusion

After comparing the values of flow velocity using two different methods, it was concluded that the bernoulli's equation serves as a good approximation to the real fluids, since the velocity calculated from the bernoulli's equation resembled to that of calculated using the manual method. However, for more accurate results, other factors such as Head loss and viscosity should be taken into consideration.

References

[1] S. Jaber and S. Jaber, "How to calculate major head loss in pipes and ducts," SimScale, Nov. 14, 2024. Available: https://www.simscale.com/blog/how-to-calculate-major-head-loss-in-pipes-and-ducts

[2] Yunus A. and John M., *Fluid Mechanics: Fundamentals and Applications*, McGraw Hill, 4 th edition, 2018.

[3] Classroom slides from Prof. Uddipta and Prof. Dilip